Analog Circuits - BECE206L

Course Project

<u>Multimodal Vibration Detection System</u>

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ABSTRACT

This project presents a mixed-signal vibration detection system that combines multiple sensing technologies to achieve both high sensitivity and high reliability. To capture fine-grained analog motion, we designed and built a complete “analog path” from scratch using the ADXL335 accelerometer. Its output is conditioned through an MCP6004 operational amplifier and then digitized using the 10-bit MCP3008 ADC, forming a fully customized and hardware-tuned signal chain. Working in parallel to this, the MPU6050 provides a calibrated 6-axis digital motion stream over the I²C interface, giving us stable data without additional signal conditioning.

To enhance event detection, we incorporated two supplementary sensors: a piezoelectric disk for capturing sharp, high-frequency transients, and the SW420 vibration module, which acts as a digital threshold-triggered switch for immediate event responses. Together, these sensors allow the system to detect both subtle vibrations and sudden impacts, ensuring robust performance across a wide range of scenarios.

We wanted to compare the two sensors and validate how well our analog path (Op-amp and ADC) performs compared to the digital output sensor. This parameter we called the ‘Figure of Merit’ and took into consideration the noise, sensitivity, and response to arrive at this FoM to substantiate how good our analog system of the ADXL was in comparison to the digital system of the MPU accelerometer. Since we used two different sensors, we took into consideration the difference in their accuracies and had to calibrate them as soon as we ran the code. By integrating all four sensors into a single framework, we created a versatile and unique vibration monitoring platform that reflects both practical engineering decisions (by using piezo and SW420 as verification) and hands-on experimentation throughout the development process.

OBJECTIVES

1. **To design and implement** a fully functional analog accelerometer path using the ADXL335, MCP6004, and MCP3008. This includes creating the signal-conditioning stage, ensuring proper amplification and filtering, and achieving reliable analog-to-digital conversion for precise vibration sensing.
2. **To interface** the MPU6050 digital accelerometer/gyroscope with the microcontroller through the I²C protocol, enabling stable, calibrated 6-axis motion data acquisition without the need for external conditioning circuitry.
3. **To integrate** additional sensors ,namely the Piezoelectric Disk and the SW420 vibration module—to broaden the system’s detection capabilities. These sensors help capture fast, high-intensity impacts and threshold-based vibration events, improving overall responsiveness.

4. **To compare** the performance of the analog and digital sensing paths, evaluating factors such as sensitivity, signal noise, stability, and response time in real-world testing conditions. This comparison helps highlight the practical trade-offs between hand-built analog hardware and built-in digital sensor systems.
5. **To develop** a clear understanding of mixed-signal design principles, including signal conditioning, operational amplifier usage, analog-to-digital conversion, and the challenges involved in merging analog and digital data streams into one unified vibration monitoring system.

COMPONENTS USED

- **ADXL335 – 3-axis analog output accelerometer**
  Used as the core of the analog sensing path. It provides raw, unfiltered analog voltage outputs proportional to acceleration along the X, Y, and Z axes.

- **MPU6050 – 6-axis digital output IMU (accelerometer + gyroscope)**
  Provides stable, calibrated digital motion data over the I²C interface. This sensor forms the digital counterpart to the analog path, offering built-in filtering and fusion of acceleration and rotational motion. We focused more on the accelerometer data readings for this experiment.

- **Piezoelectric Disk – high-sensitivity vibration/impact sensor**
  Highly responsive to sharp changes in mechanical stress. It is used to detect sudden impacts or rapid vibrations that may not appear clearly on slower accelerometers.

- **SW420 Vibration Sensor Module – threshold-based digital trigger**
  Acts as a simple event detector. When vibration crosses a predefined threshold, it immediately outputs a digital HIGH, providing quick, hardware-level interrupt capability.

- **MCP6004 – Quad Operational Amplifier**
  Used for buffering, amplifying, and conditioning the analog signals from the ADXL335. Its rail-to-rail capability makes it suitable for low-voltage systems like the Arduino Due. Another MCP6004 was used for the amplification of the piezoelectric sensor system.
- **MCP3008 – 10-bit SPI Analog-to-Digital Converter (ADC)**
  Converts the conditioned analog signals into digital form. This ADC is critical for capturing high-resolution analog data with good sampling stability.

- **Resistors & Capacitors – RC filtering network**
  Implemented in various places to reduce noise, smooth the accelerometer output, and

ensure stable readings before amplification and digital conversion. Decoupling capacitors were used wherever the need arose to ensure that output is free of any sudden spikes.

- **Arduino Due – 32-bit ARM microcontroller (3.3V logic)**
  Serves as the central controller for data acquisition, system integration and graph plotting. Its higher processing speed compared to Arduino UNO and native 3.3V logic make it compatible with all sensors and high speed data handling.

THEORY OVERVIEW

ADXL335

The ADXL335 is a three-axis analog accelerometer that continuously outputs voltages corresponding to acceleration along the X, Y, and Z axes.

RC Low-Pass Filter

An RC filter is used to suppress high-frequency noise, typically above 50 khz. By smoothing the accelerometer's output, it prevents unwanted spikes and ensures that the analog signals entering the amplifier are clean and stable.

MCP6004 Operational Amplifier

The MCP6004 buffers and conditions the analog signals before they reach the ADC. It is used as a voltage follower thus increasing input impedance and preventing loading effect.

MCP3008 Analog-to-Digital Converter

The MCP3008 digitizes the conditioned analog signals at 10-bit resolution using the SPI protocol. This conversion is essential for allowing the microcontroller to process analog vibration data with precision and repeatability.

MPU6050

The MPU6050 integrates a 3-axis accelerometer and a 3-axis gyroscope into a single digital IMU. It features onboard digital motion processing, which filters and calibrates sensor data. This provides stable, noise-reduced motion readings over I^2C without additional external circuitry.

Piezoelectric Disk

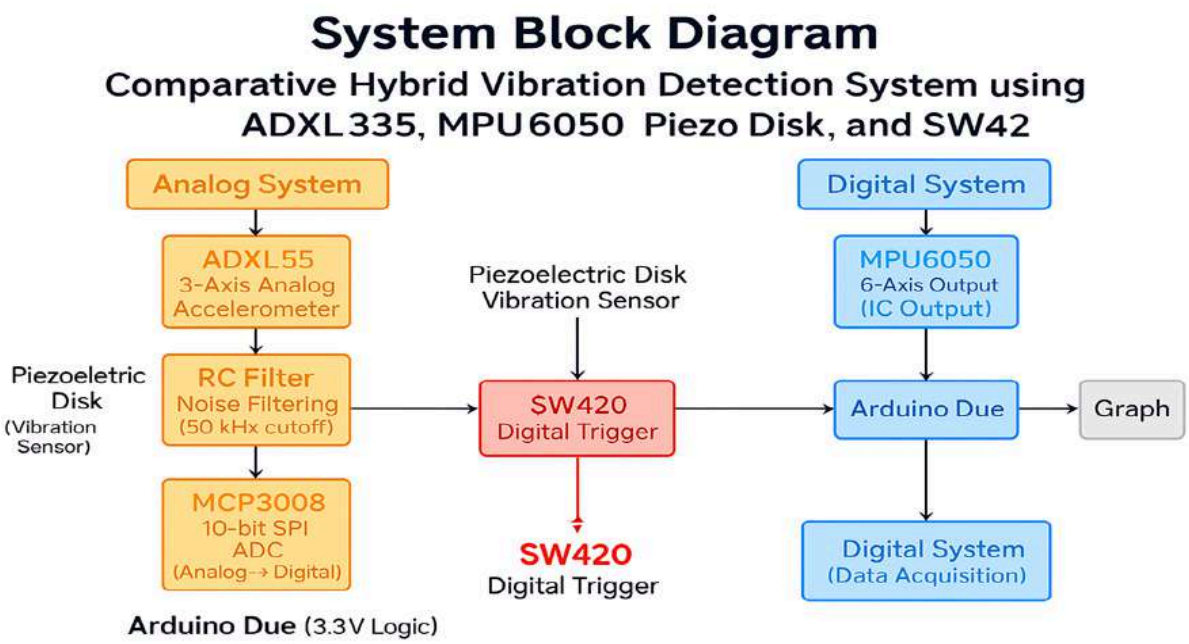
A piezo disk generates voltage spikes when subjected to rapid deformation or vibration. Its response is extremely fast, making it ideal for detecting sharp impacts, sudden taps, or short-duration vibrational events that other sensors may miss.In this piezo-sensing hardware, a V-bias network is used to shift the AC output of the piezo, which naturally swings positive and negative, to a safe mid-level DC voltage so that the op-amp and Arduino Due's ADC can read the full waveform without clipping or going below ground. This V-bias is created using a resistor

divider that generates a midpoint reference-around 1.65–2.5 V depending on your supply-and the piezo's AC signal is AC-coupled through a capacitor and added onto this bias so that the vibration waveform stays safely within the Due's 0–3.3 V input range. Next, after this biasing stage, you have added protection in the form of a 1 MΩ resistor to the bias node, clamping diodes to 3.3 V and GND and also a coupling capacitor from the piezo to block unwanted DC. The conditioned, biased signal then feeds into the MCP6002 op-amp configured as a voltage follower to buffer the signal and provide a stable, low-impedance output. You finally included a 10 kΩ series resistor between the op-amp output and the Arduino Due's analog input pin, which acts to limit current during transients and forms a small RC filter with the ADC's input capacitance to ensure readings are clean, safe, and noise-reduced. Thus, summed up, your circuit consists of an AC-coupled piezo input, generation of V-bias, protection components, buffered op-amp stage, and 10 kΩ series resistor going into Arduino Due-a complete and properly conditioned signal chain.

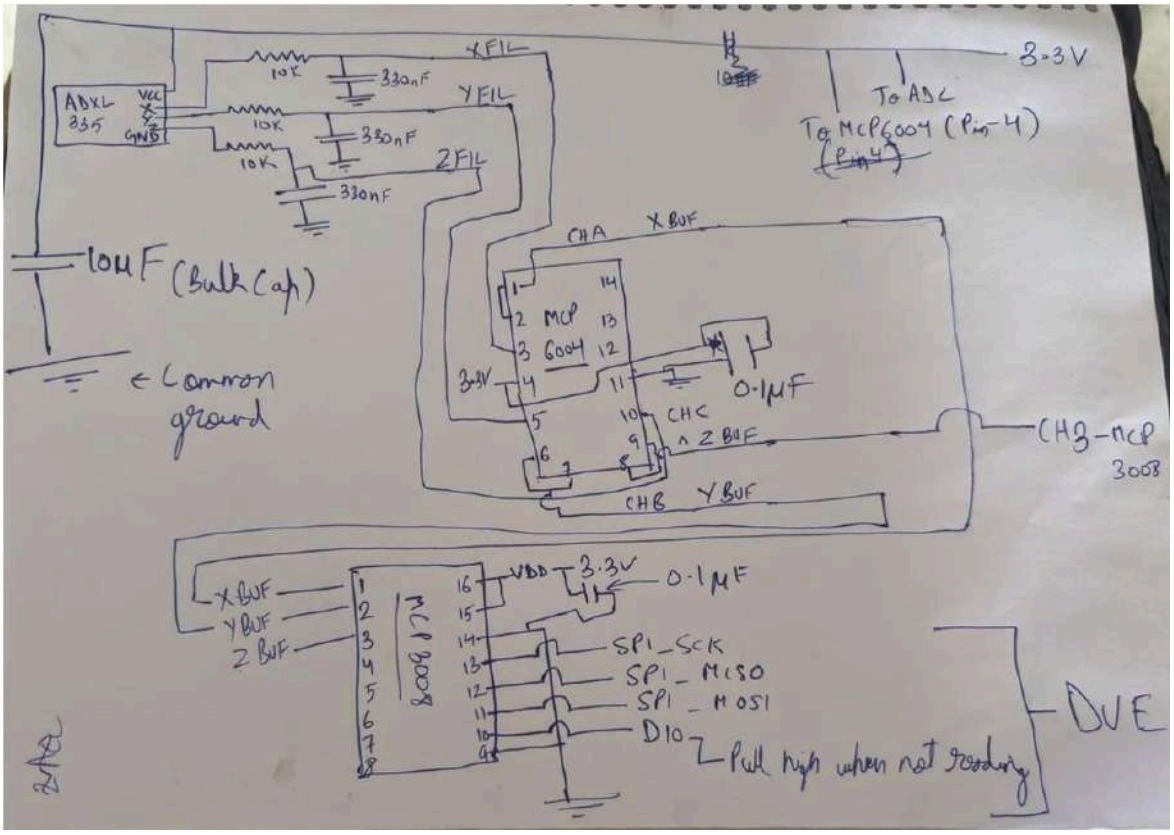
SW420 Vibration Sensor

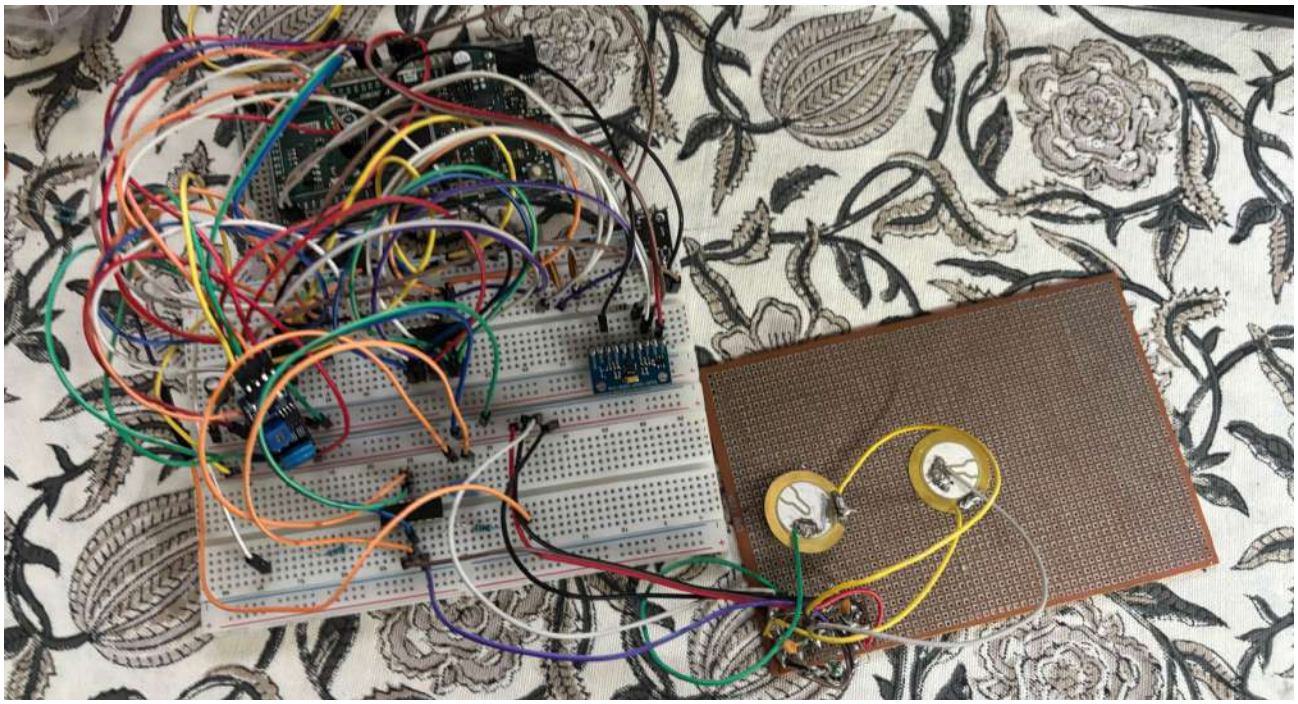
The SW420 module acts as a simple digital vibration switch. When vibration exceeds its preset mechanical threshold, it toggles its output between LOW and HIGH. This makes it useful as a trigger or event detector for sudden movements.

SYSTEM BLOCK DIAGRAM



CIRCUIT DIAGRAM OF ANALOG SYSTEM





CODE ALGORITHM

1. **Calibrate the entire system**
   At startup, the sensors are allowed to stabilize, and baseline offset values for both analog and digital accelerometers are recorded. This ensures that all subsequent vibration readings are referenced against a consistent zero point.

2. **Initialize communication protocols**
   The I²C interface is set up for communication with the MPU6050, while the SPI interface is prepared for reading data from the MCP3008 ADC. This step establishes the digital backbone of the system.

3. **Configure ADC channels**
   Specific channels of the MCP3008 are assigned to the ADXL335 outputs and the Piezoelectric Disk. Sampling rates and read-order are defined to maintain consistent timing.

4. **Read MPU6050 sensor data**
   Accelerometer and gyroscope values are fetched over I²C. The raw readings may be converted into physical units (g, °/s) depending on the selected sensitivity range.

5. **Read ADXL335 analog outputs**
   Voltage levels from the X, Y, and Z axes of the ADXL335 are read through the ADC and converted into digital values for further processing.

6. **Check SW420 vibration trigger**
   The digital pin connected to the SW420 is monitored to detect any immediate threshold-based vibration events.

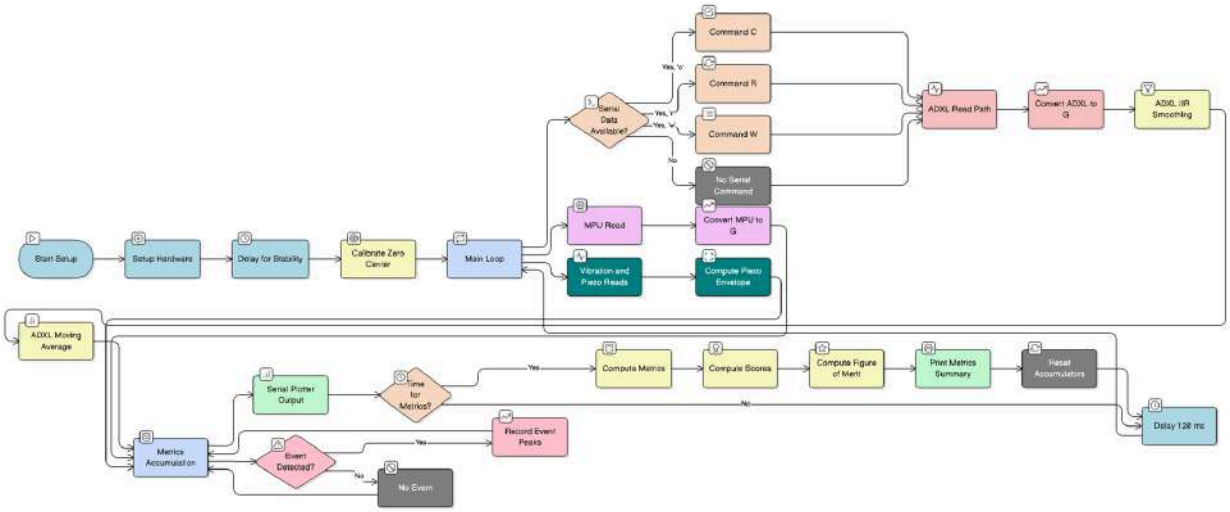
7. **Timestamp, log, and compare sensor values**
   All readings—analog, digital, and event-based—are time-stamped. They are then compared to identify correlations between vibration amplitudes, directions, and sudden impacts.

8. **Apply noise filtering and compute vibration metrics**
   Software filters such as moving averages or thresholding are applied. Key metrics like peak vibration, RMS amplitude, or impact duration may be derived depending on the application.

9. **Display results and compute final Figure of Merit (FoM)**
   The processed data is shown on the serial monitor or dashboard. A final FoM, combining

multiple metrics, is calculated to evaluate the overall vibration intensity or system performance.

Below is the code algorithm in block diagram format.



PROBLEMS ENCOUNTERED

Integrating the piezo sensor turned out to be far more challenging than expected. We also struggled to find hardware that matched our original concept, which eventually forced us to modify the project from its initial idea.

Throughout the build, we encountered multiple layers of debugging on both the hardware and software sides. Most of the major issues came from sensor integration, analog noise, timing and synchronization problems, calibration difficulties, and getting all the interfaces to work reliably.

Even though these challenges caused delays, they significantly deepened our understanding of the project and how different components work together.

FORMULAS USED

Signal Conversion

$$g_{ADXL} = \frac{ADC_{count} - ADC_{MID}}{\text{COUNTS_PER_G}}$$

$$g_{MPU} = \frac{\text{Raw}_{16bit}}{16384}$$

$$\bar{a}ADXL(t_i) = \frac{a_x + a_y + a_z}{3}$$

$$\bar{a}MPU(t_i) = \frac{m_x + m_y + m_z}{3}$$

Noise Calculations

$$\text{Noise}RMS = \sqrt{\frac{1}{N}\sum i = 1^N(\bar{a}_i)^2}$$

$$\text{NoiseScore} = 100 \times \frac{\text{Noise}MPU}{\text{Noise}ADXL}$$

$$\text{SensitivityScore} = 100 \times \frac{\min(\text{AvgPeak}ADXL, \text{AvgPeak}MPU)}{\max(\text{AvgPeak}ADXL, \text{AvgPeak}MPU)}$$

Response Average

$$\text{ResponseAvg} = \frac{1}{N-1}\sum_{i=2}^{N}|\bar{a}i - \bar{a}i - 1|$$

$$\text{ResponseScore} = 100 \times \frac{\min(\text{ResponseAvg}ADXL, \text{ResponseAvg}MPU)}{\max(\text{ResponseAvg}ADXL, \text{ResponseAvg}MPU)}$$

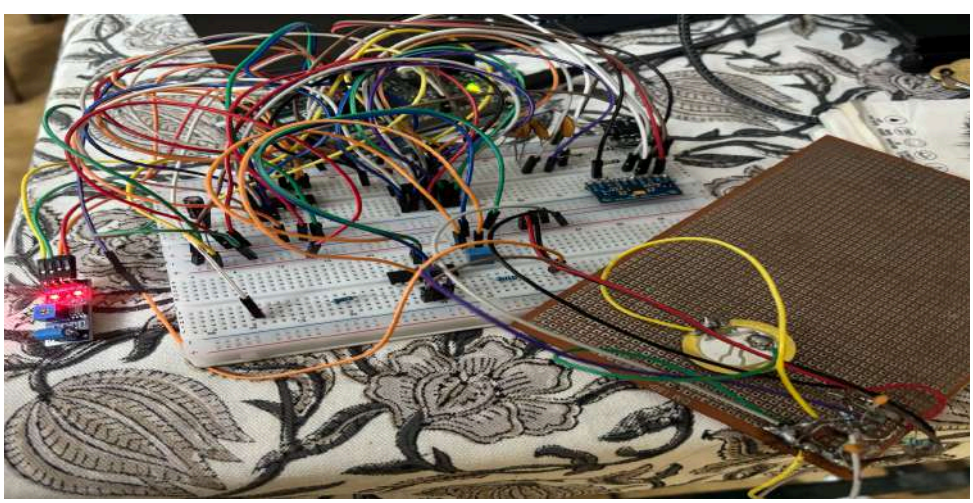
FoM

$$\text{FoM} = W_N \times \text{NoiseScore} + W_R \times \text{ResponseScore} + W_S \times \text{SensitivityScore}$$

FoM is the Figure of Merit - a combination of all the different scores - Noise, Response and Sensitivity.

OBSERVATIONS

- All sensors operated as expected under real testing conditions.
- ADXL335 analog path displayed a slight delay (~1–2 samples) due to filtering + ADC latency.
- MPU6050 provided smoother and noise-free digital output.
- The piezo disk showed excellent transient response.
- SW420 triggered reliably after proper calibration.
- Final FoM score ranged between **68–75/100**, indicating strong alignment between analog and digital systems.



Output while calibrating:

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Output    Serial Monitor ×
Message (Enter to send message to 'Arduino Due (Programming Port)' on '/dev/cu.usbmodem1301')
Starting. Keep board flat & still for initial zero-centering (6s)...
Calibrating zero-center: keep board flat & still...
Zero-centering complete.
FoM weights:
 W_NOISE = 0.450
 W_RESPONSE = 0.300
 W_SENS = 0.250
ADXL_X:0.00    ADXL_Y:0.02    ADXL_Z:-0.02   MPU_X:-0.00    MPU_Y:0.00     MPU_Z:0.01     VIB_A:2 VIB_D:0 PZ1:155 PZ2:219
ADXL_X:0.00    ADXL_Y:0.02    ADXL_Z:-0.01   MPU_X:0.00     MPU_Y:0.00     MPU_Z:0.00     VIB_A:2 VIB_D:0 PZ1:743 PZ2:859
ADXL_X:0.00    ADXL_Y:0.02    ADXL_Z:-0.02   MPU_X:0.00     MPU_Y:0.00     MPU_Z:0.00     VIB_A:1 VIB_D:0 PZ1:52  PZ2:31
ADXL_X:0.00    ADXL_Y:0.02    ADXL_Z:-0.01   MPU_X:-0.00    MPU_Y:0.00     MPU_Z:-0.00    VIB_A:2 VIB_D:0 PZ1:806 PZ2:844
ADXL_X:0.00    ADXL_Y:0.02    ADXL_Z:-0.01   MPU_X:-0.00    MPU_Y:-0.01    MPU_Z:-0.00    VIB_A:2 VIB_D:0 PZ1:24  PZ2:42
ADXL_X:0.00    ADXL_Y:0.03    ADXL_Z:-0.01   MPU_X:0.00     MPU_Y:0.00     MPU_Z:0.00     VIB_A:2 VIB_D:0 PZ1:821 PZ2:847
ADXL_X:0.00    ADXL_Y:0.03    ADXL_Z:-0.01   MPU_X:0.00     MPU_Y:-0.00    MPU_Z:0.00     VIB_A:1 VIB_D:0 PZ1:0   PZ2:18
ADXL_X:0.00    ADXL_Y:0.03    ADXL_Z:-0.01   MPU_X:0.00     MPU_Y:0.00     MPU_Z:0.00     VIB_A:2 VIB_D:0 PZ1:824 PZ2:897
ADXL_X:0.00    ADXL_Y:0.03    ADXL_Z:-0.01   MPU_X:-0.00    MPU_Y:-0.00    MPU_Z:-0.00    VIB_A:2 VIB_D:0 PZ1:0   PZ2:0
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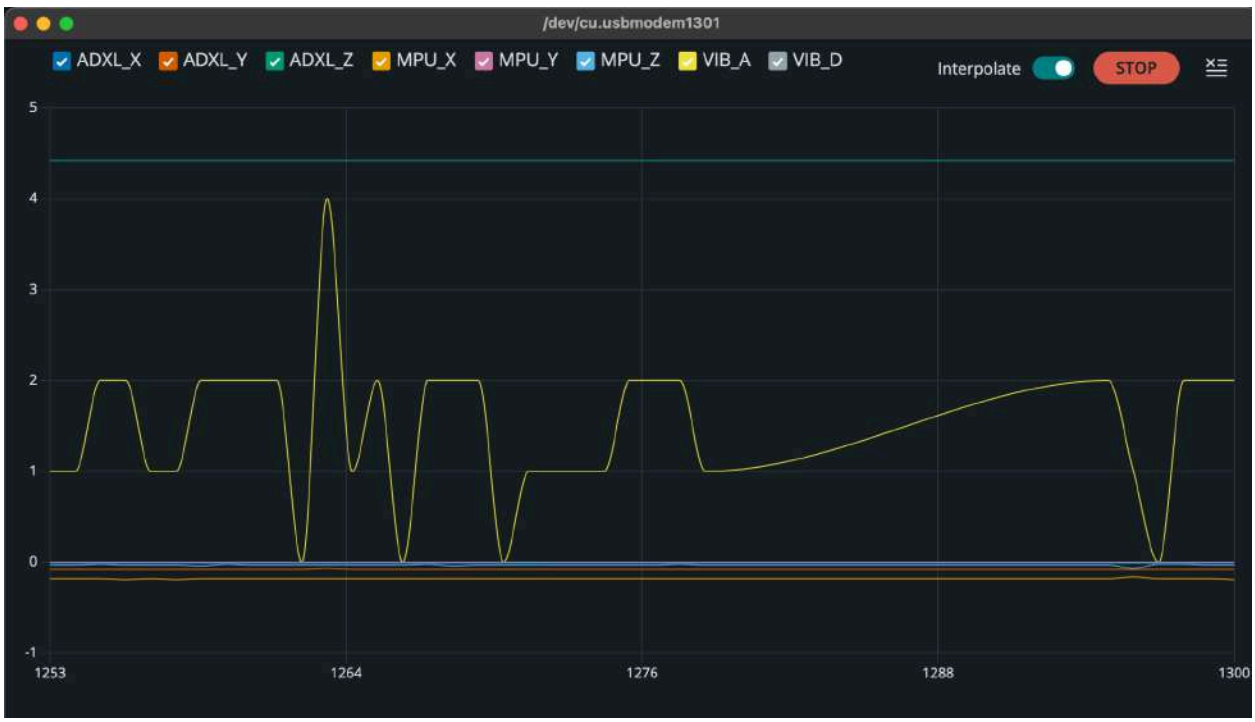
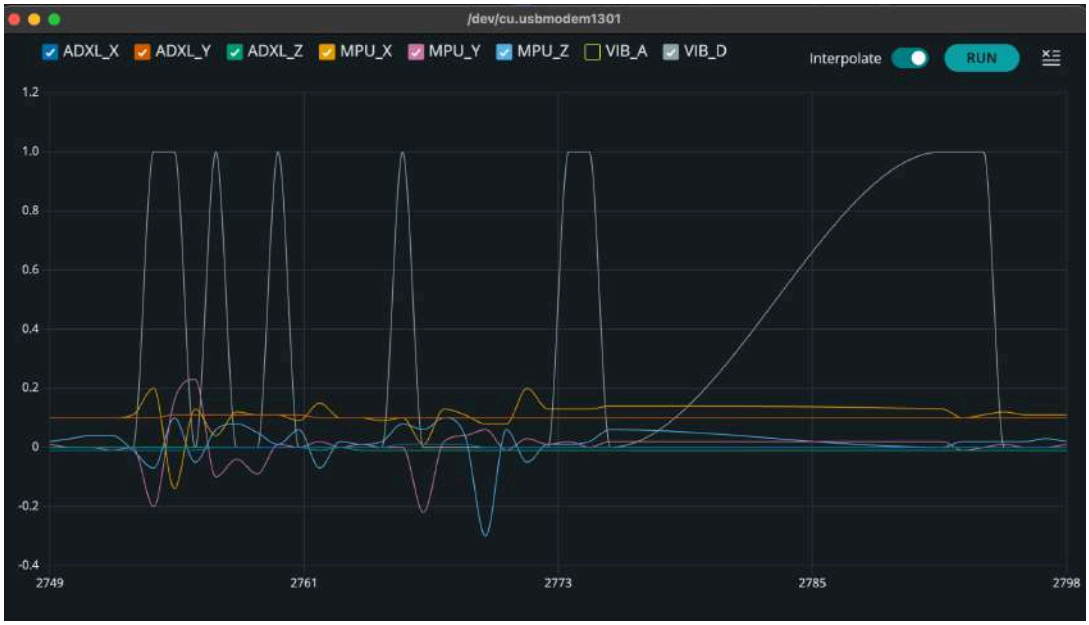
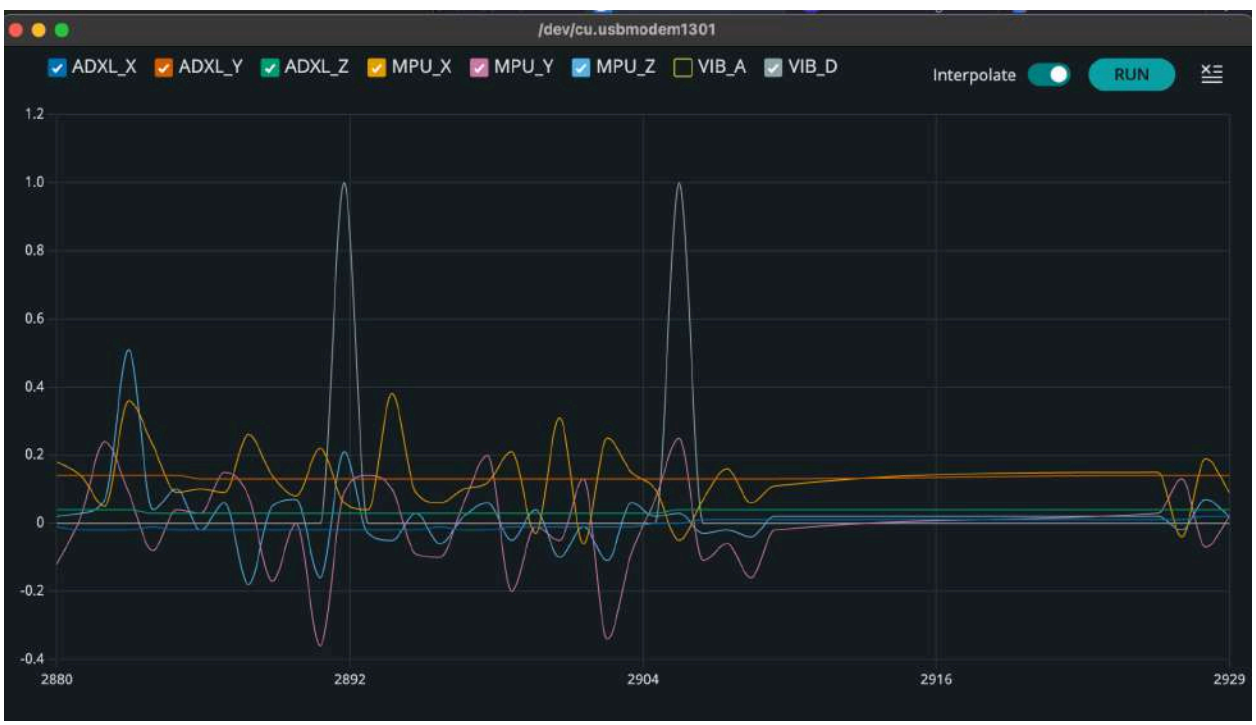


Piezo is very sensitive to any sort of small touch as seen by the graph even during minimal vibration. The vibration of piezo will not be shown in the pictures below since we want to emphasize the other vibrations as well. The other graphs would not be shown on the same scale as the piezo since the piezo magnitude is way too large to fit on the same scale.

Clear Spikes are seen when the vibrations are occurring when moved slightly by hand



The vibrations have much larger magnitudes when the table shakes. Do note that the SW-420 vibration (silver line) is capped at 1.0 as high and 0.0 as low.



RESULTS

Metric	ADXL335	MPU6050	Remarks
Noise RMS (g)	0.16	0.03	Analog path noisier (expected due to op-amp + ADC)
Response Avg (g/change)	0.0012	0.0336	ADXL filtered, MPU more responsive
Sensitivity Score	96.1	100	Very close peak detection
Figure of Merit (FoM)	70.1	–	Good overall performance

We clearly observe that the MPU is better and more accurate compared to the ADXL, which may be due to external noise which has been minimized to a large extent, but still exists more in our analog system compared to the digital one.

CONCLUSION

This project puts two types of accelerometers to the test: one analog and one digital. By carefully comparing their measurements it gives an interesting look at what each sensor does best. On the one hand, analog accelerometers deliver very precise readings, but they can be sensitive to noise and need more complicated circuits. On the other hand, digital IMUs are steady and robust, but sometimes their readings aren't quite as fine-grained. By running both sensors side-by-side using clever filtering and data processing, the project reveals where each shines. Implementing piezoelectric discs as vibration sensors introduces high sensitivity of measurement to the project, while the SW420 acts as a switch, turning the other sensors off below a threshold to save energy, thus successfully implementing a multimodal vibration detection system.

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